

Indoor Air Quality Monitoring System

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Abstract—Indoor Air Quality (IAQ) is a critical determinant of human health and productivity, which is often overlooked in favor of outdoor pollution monitoring. This paper presents the design and implementation of a low-cost, scalable Indoor Air Quality Monitoring System based on the Internet of Things (IoT). The system utilizes an ESP32 Dev Board microcontroller integrated with DHT22, MQ135, and PMS5003 sensors to measure temperature, humidity, CO_2 levels, and particulate matter (PM2.5/PM10) in real-time. Data is transmitted via MQTT to AWS IoT Core and archived in Amazon S3 for historical analysis using Amazon Athena. Computed metrics such as Air Quality Index (AQI), Heat Index, and Dew Point are calculated at the node level to enhance payload value. An actuator system provides immediate local feedback via LEDs and buzzers when hazardous thresholds are breached. The system demonstrates a robust architecture for continuous environmental monitoring with visualization capabilities through a dashboard implemented using Streamlit.

I. INTRODUCTION

A. Background

The Indoor air pollution is a major contributor to respiratory illnesses, reduced cognitive performance, and long-term cardiovascular risks [1]. The World Health Organization reports that indoor pollution results in millions of premature deaths annually. Since individuals spend nearly 90% of their time indoors, pollutants such as volatile organic compounds (VOCs), particulate matter (PM), and elevated concentrations of carbon dioxide (CO_2) accumulate in enclosed spaces and pose severe health concerns. In rapidly urbanizing cities such as Karachi, outdoor pollution often infiltrates indoor environments, making continuous indoor air monitoring essential for ensuring occupant well-being.

B. Existing Solutions

Several commercial and research-based Indoor Air Quality (IAQ) monitoring systems exist, but most are either expensive or closed-source. Research-grade particulate monitors such as the PMS5003 have been widely used in environmental sensing studies due to their high responsiveness and reliability in measuring PM2.5 and PM10 concentrations [2]. For carbon dioxide sensing, prior work shows that NDIR-based sensors such as the SCD30 offer significantly higher accuracy compared to low-cost chemical sensors, making them a preferred choice in validated IAQ systems [3].

In IoT-enabled monitoring platforms, cloud services such as AWS IoT Core are commonly used for secure device connectivity, MQTT messaging, and scalable data storage,

forming the backbone of many modern sensing architectures [4]. These existing systems highlight the effectiveness of combining low-cost sensors with cloud analytics, but also emphasize the need for open, customizable, and affordable IAQ monitoring solutions.

C. Project Objectives

The objective of this project is to design an Indoor Air Quality Monitoring System that leverages IoT and cloud technologies to provide actionable environmental insights. Specifically, the project aims to:

- 1) To design a sensor node capable of measuring key environmental parameters: Temperature, Humidity, CO_2 , and Particulate Matter that can help assess Air Quality.
- 2) To establish a reliable IoT-cloud data pipeline using AWS services (IoT Core, S3, Athena) for data acquisition, storage, and analysis.
- 3) To integrate local actuation for immediate hazard alerts.

II. METHODOLOGY

A. IoT Layered Architecture

The design of the system can be described using the four-layer IoT architecture model, consisting of the *Perception* layer for sensing, *Transport* layer for communication, *Processing* layer for data management, and *Application* layer for end-user interaction, all integrated into a cohesive system.

1) *Perception Layer*: The Perception Layer comprises the physical sensing node built around an ESP32 Dev Board. It is responsible for acquiring environmental parameters and performing lightweight computations before data transmission. Three sensors constitute the node design:

- **DHT22**: A digital sensor used for measuring ambient temperature and relative humidity. It provides a reliable baseline for thermal comfort calculations.
- **MQ135**: A gas sensor utilized for detecting various gases like Benzene, Alcohol, and Smoke. In this implementation, however, it has been calibrated to monitor CO_2 concentration levels (ppm) only.
- **PMS5003**: A laser scattering particle sensor that measures the concentration of suspended particles. It provides precise data for PM1.0, PM2.5, and PM10 concentrations in $\mu g/m^3$.

The ESP32 periodically samples each sensor, computes derived metrics such as AQI, Heat Index, and Dew Point,

and formats all readings into a JSON payload for transmission.

```
latest_data = {
  "device_id": None,
  "ip_address": None,
  "temperature_C": None,
  "humidity_percent": None,
  "dew_point_C": None,
  "heat_index_C": None,
  "pms_ok": None,
  "pm1_0_ugm3": None,
  "pm2_5_ugm3": None,
  "pm10_ugm3": None,
  "aqi_index": None,
  "aqi_category": None,
  "co2_ppm": None,
  "timestamp_ms": None
}
```

2) *Transport Layer*: The Transport Layer enables wireless communication between the sensing node and cloud infrastructure. Two parallel communication pipelines were implemented:

- **Primary Path (MQTT over Wi-Fi)**: The ESP32 publishes sensor readings to AWS IoT Core using MQTT secured through TLS (Topic: `esp32/esp32-to-aws`). MQTT was selected for its low bandwidth requirements and suitability for real-time monitoring.
- **Secondary Path (HTTP over Wi-Fi)**: A secondary pipeline was implemented to send the same JSON payload to a Flask server hosted on an Amazon EC2 instance via HTTPS POST requests. The purpose of this channel was to observe the differences in the operations of the HTTPS and MQTT protocols.

The MQTT pipeline is utilized for the primary control loop and long-term storage.

3) *Processing Layer*: The Processing Layer resides entirely on AWS and is responsible for persistent storage, data querying, and cloud-side actuation.

- **AWS IoT Core Rule Engine**: Routes incoming MQTT messages to an Amazon S3 bucket.
- **Amazon S3**: Stores incoming JSON payloads using timestamp-based object keys (partitioning strategy) for efficient retrieval.
- **Amazon Athena**: Performs SQL queries directly over S3 data using a JSON SerDe, enabling analytics without the need for a dedicated database server.
- **AWS Lambda (Control Plane)**: Configured to handle downlink communication. When triggered by the dashboard API, the Lambda function publishes command payloads (e.g., `{"buzzer": "ON"}`) to the `esp32/esp32-to-aws` topic, enabling remote actuation.

4) *Application Layer*: The Application Layer provides visualization capabilities for end users. A dashboard was developed using **Streamlit**, which connects to Amazon Athena to retrieve real-time data. The dashboard displays:

- Temperature and humidity trends,

- Particulate matter levels (PM2.5/PM10),
- AQI classification and color-coded severity

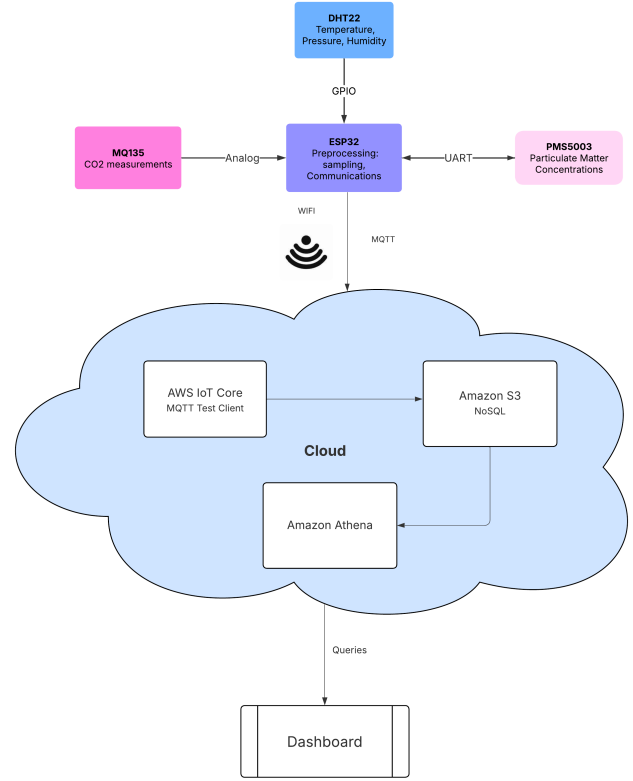


Fig. 1. System-Level IoT Architecture

B. Computed Metrics at the Edge

To enrich the transmitted payload and perform lightweight edge analytics, environmental metrics relevant to Air Quality are computed directly on the ESP32.

1) *Heat Index (HI)*: The Heat Index combines air temperature and relative humidity to determine the human-perceived equivalent temperature. The Rothfus regression equation provided by the U.S. National Weather Service [7] is used:

$$HI = -42.379 + 2.049T + 10.143R - 0.224TR - 6.83 \times 10^{-3}T^2 - 5.48 \times 10^{-2}R^2 + \dots \quad (1)$$

where T is the temperature in Fahrenheit and R is the relative humidity.

2) *Dew Point*: Dew point is calculated to estimate the moisture content in the air, providing insight into potential condensation issues (such as mold growth risk). This dew point is calculated using a simpler approximation commonly referenced in meteorological studies [9]:

$$T_{dp} = T - \frac{100 - RH}{5} \quad (2)$$

3) *Air Quality Index (AQI)*: The AQI is calculated based on PM2.5 concentrations using the piecewise linear interpolation formula standard in environmental monitoring [8]:

$$I = \frac{I_{high} - I_{low}}{C_{high} - C_{low}}(C - C_{low}) + I_{low} \quad (3)$$

where C is the measured concentration, and I represents the index boundaries for the specific pollution category. This equation is not directly implemented; rather, a pre-computed look-up table is used instead.

C. Actuator Logic and Verification

To validate the local actuation functionality, a hybrid logic system was implemented. The ESP32 compares readings against preset safety thresholds while listening for cloud overrides:

$$Action = (AQI > 150) \text{ or } (CO_2 > 1000) \quad (4)$$

To verify, a test case has been designed which involves intentionally increasing the particulate concentration near the sensor node. As PM2.5 levels rise, the calculated AQI will exceed the *Unhealthy* category, activating the LED and buzzer system. This confirms the correct operation of the sensing–computation–actuation loop.

III. DISCUSSION

The development of the end-to-end IoT pipeline provided a valuable opportunity to critically examine the engineering decisions, data flow architecture, and system behaviour across each stage of implementation.

The completed pipeline demonstrates reliable end-to-end data transmission. Each step of the process involved several design choices for communication, API handling, and data structuring in accordance with the project’s intended functionality. Sensor readings could be successfully acquired, encoded, transmitted, logged, and visualised, showing that the architectural flow from device to dashboard operated coherently. The MQTT-to-HTTP handoff, cloud API ingestion, and frontend rendering all work without major failures, indicating that the system design is technically sound..

A. Limitations and Challenges

A major limitation of the node relates to environmental variability. Without measuring performance across different indoor environments, interference scenarios, or device states, the system’s resilience remains uncertain. The lack of stress testing also limits conclusions about scalability beyond a single-node setup.

Additionally, the project can be expanded by deploying multiple nodes for data collection on the cloud, allowing for analytics to be performed to obtain insights about the indoor environment.

B. Reflection on the Engineering Process

This project provided a meaningful experience into IoT development. The justification behind design decisions, such as the choice of MQTT, the JSON schema, or the dashboard architecture became clearer only through hands-on implementation. The project provided a foundational base which allows for exploration into automated testing tools, simulation environments, and containerised deployment strategies to improve reliability and maintainability.

IV. CONCLUSION

This project successfully demonstrates a scalable, end-to-end IoT solution for Indoor Air Quality monitoring. By combining edge computing (ESP32) with cloud services (AWS), we achieved a system that is both responsive locally and capable of long-term data analysis. The integration of derived metrics like the Heat Index and AQI adds significant context to raw sensor data, making it meaningful for end-users. Future work can focus on integrating a more precise NDIR CO_2 sensor (e.g., SCD30), replicating the setup with multiple nodes to gather a large number of data points from varied indoor environments, permitting the implementation of machine learning models on AWS to predict air quality trends.

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